

Hands-on 1 Classification

(Keras3/Tensorflow environnement)



— Montreal, June 15–19 —

DLMI2026

Thomas Grenier / Michaël Sdika / Valentine Wagnier-Dauchelle

Fonds de recherche
Nature et
technologies
Québec 



Preliminary remarks

- All four hands-on can be done on a computing cloud such as that of *CalculQuebec*, *Google Colab*, ... or on your own computer
- **For beginners** : we recommend you to start with *CalculQuebec*
- For advanced users with CPU/GPU access
 - You can run on your specific hardware
 - Tips for installing the required environnement are [here](#).

First, setup your *CalculQuebec* Account

(cf. DLMI 2026 hands on setup instructions)

1. *Create an account on CalculQuebec Cloud and log-in*
<https://jupyter.dmlri.calculquebec.cloud/hub/login>
2. Create a compute node with
 1. Memory at 4096MB
 2. GPU 1x1G.5GB
3. Start the jupyter-lab server and wait for it **Steps covered here**
4. Download the hands-on material into your jupyter-lab
5. Open a notebook
6. Cells philosophy and execution

Jupyter Lab interface (can change)

The screenshot shows the Jupyter Lab web interface. On the left is a file browser sidebar with a table of files and folders. Two orange arrows point to the file browser: one points to the 'upload' button (a square with an upward arrow) and the other points to the file list area. A red text box at the bottom left says "Drag and drop files here (or via the upload button)". The main area is titled "Launcher" and contains several sections of application tiles. A green arrow points to the "Launcher tab" label at the top of the main area. Another green arrow points to the "Terminal" tile in the "Other" section. The "Notebook" section contains tiles for Python 3.11, LibreQDA, Mate Desktop (VNC), MLflow, OpenRefine, Python bundle, RStudio, Tensorboard, and VS Code. The "Console" section contains tiles for Python 3.11 and Python bundle. The "Other" section contains tiles for Terminal, Job statistics, Text File, Markdown File, Python File, and Show Contextual Help. The "Documentation" section contains tiles for General and Jupyter. A small notification box in the bottom right corner says "Would you like Jupyter news".

File browser sidebar:

Name	Modified
projects	last mo.
scratch	last mo.

Main area (Launcher tab):

- Notebook**
 - Python 3.11
 - LibreQDA [?]
 - Mate Desktop (VNC) [?]
 - MLflow [?]
 - OpenRefine [?]
 - Python bundle
 - RStudio [?]
 - Tensorboard [?]
 - VS Code [?]
- Console**
 - Python 3.11
 - Python bundle
- Other**
 - Terminal
 - Job statistics
 - Text File
 - Markdown File
 - Python File
 - Show Contextual Help
- Documentation**
 - General
 - Jupyter

Drag and drop files here (or via the upload button)

Launcher tab

Terminal

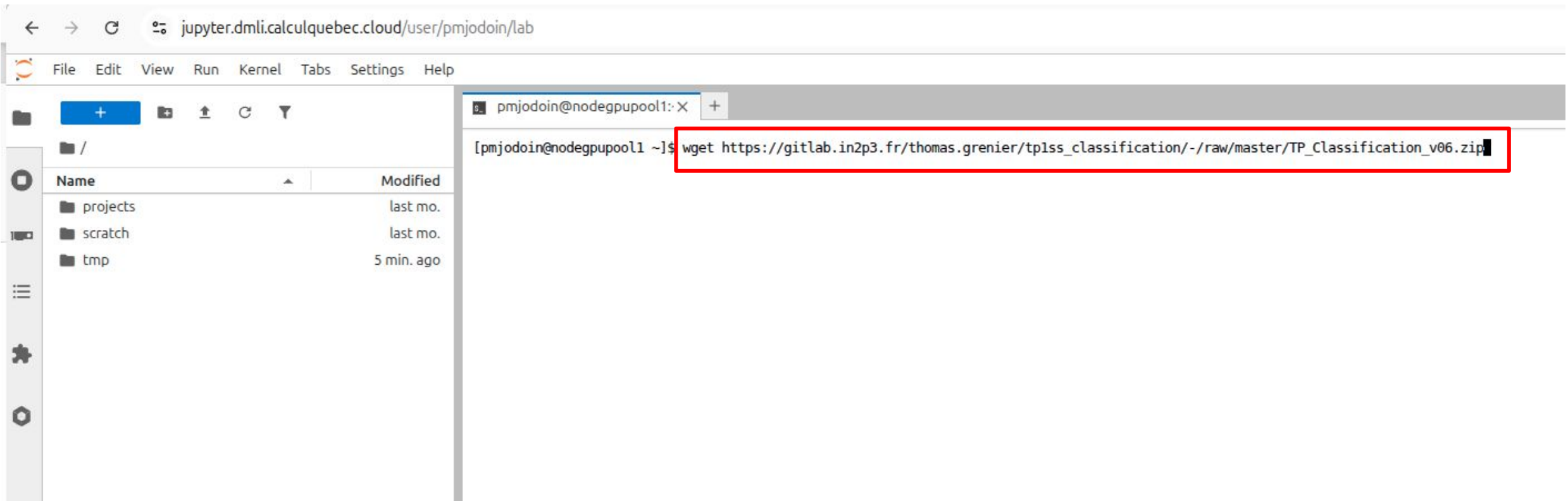
Would you like Jupyter news

Download Hands-on materials

a- Open a **Terminal (c.f. previous page)**

b- download from the terminal with the following command :

```
wget https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip
```



Download Hands-on materials

a- Open a **Terminal**

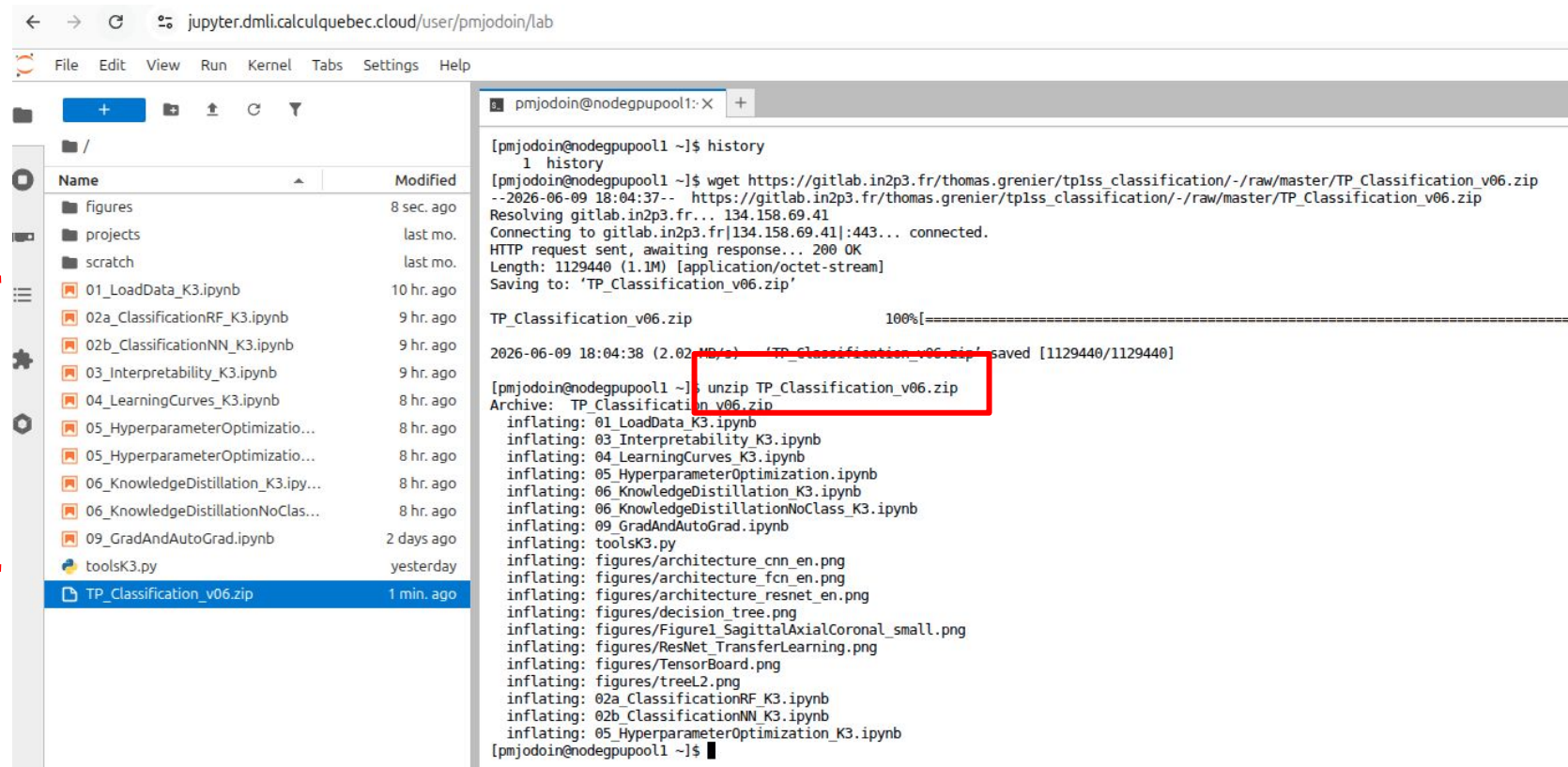
b- download from the terminal with :

```
wget https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip
```

c- unzip the file (in a Terminal)

```
unzip TP_Classification_v06.zip
```

unzipped files



```
[pmjodoin@nodegpupool1 ~]$ history
1 history
[pmjodoin@nodegpupool1 ~]$ wget https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip
--2026-06-09 18:04:37-- https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip
Resolving gitlab.in2p3.fr... 134.158.69.41
Connecting to gitlab.in2p3.fr[134.158.69.41]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1129440 (1.1M) [application/octet-stream]
Saving to: 'TP_Classification_v06.zip'

TP_Classification_v06.zip      100%[=====]
2026-06-09 18:04:38 (2.02 MB/s) 'TP_Classification_v06.zip' saved [1129440/1129440]

[pmjodoin@nodegpupool1 ~]$ unzip TP_Classification_v06.zip
Archive: TP_Classification_v06.zip
  inflating: 01_LoadData_K3.ipynb
  inflating: 03_Interpretability_K3.ipynb
  inflating: 04_LearningCurves_K3.ipynb
  inflating: 05_HyperparameterOptimization.ipynb
  inflating: 06_KnowledgeDistillation_K3.ipynb
  inflating: 06_KnowledgeDistillationNoClass_K3.ipynb
  inflating: 09_GradAndAutoGrad.ipynb
  inflating: toolsK3.py
  inflating: figures/architecture_cnn_en.png
  inflating: figures/architecture_fcnn_en.png
  inflating: figures/architecture_resnet_en.png
  inflating: figures/decision_tree.png
  inflating: figures/Figure1_SagittalAxialCoronal_small.png
  inflating: figures/ResNet_TransferLearning.png
  inflating: figures/TensorBoard.png
  inflating: figures/treeL2.png
  inflating: 02a_ClassificationRF_K3.ipynb
  inflating: 02b_ClassificationNN_K3.ipynb
  inflating: 05_HyperparameterOptimization_K3.ipynb
[pmjodoin@nodegpupool1 ~]$
```

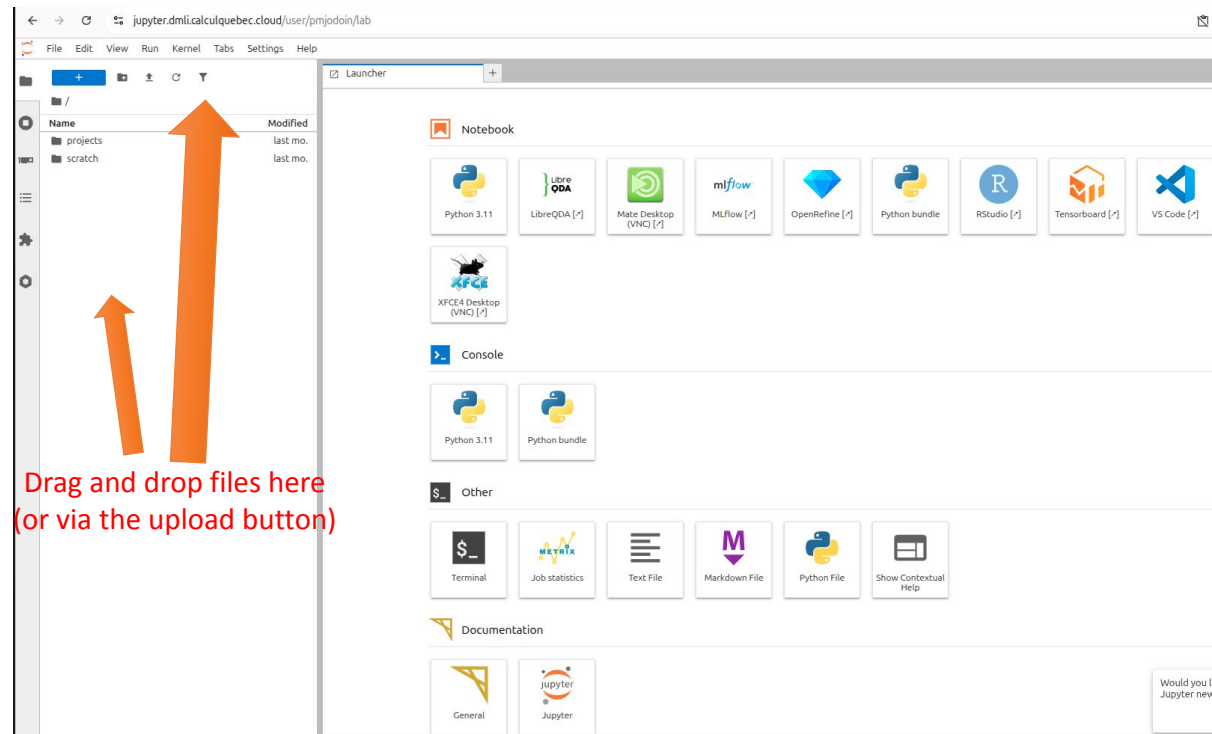
Download Hands-on materials

If the previous steps did not work, you can

a) download the zip file on your computer

https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip

b) Upload it in your online environment



Download Hands-on materials

If the previous step did not work, you can

a) download the zip file on your computer

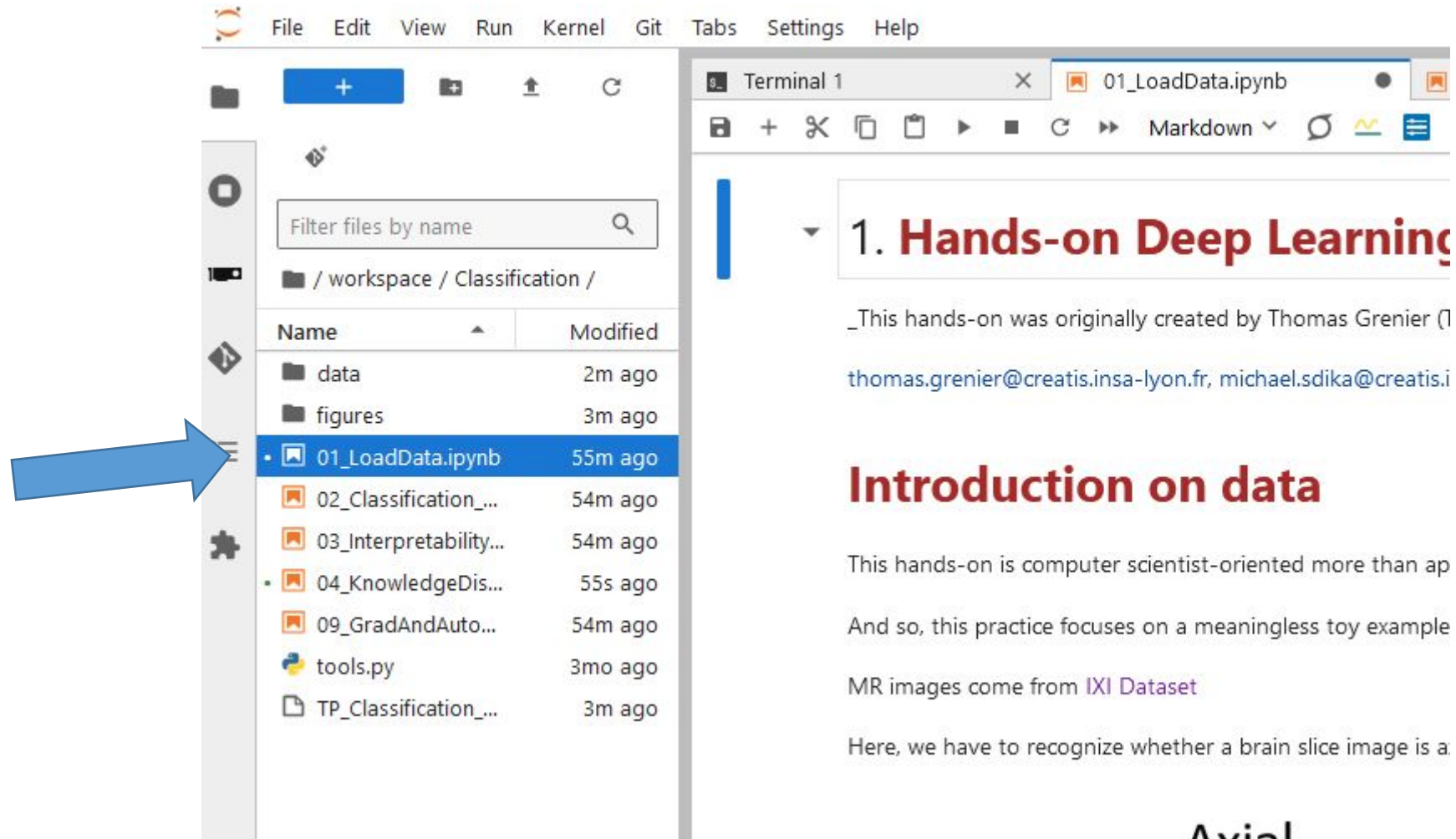
https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip

b) Upload it in your online environment

c) Unzip the file

Select and open the desired notebook

As example, double clic on : 01_LoadData.ipynb



The screenshot displays the JupyterLab interface. On the left, the file browser shows the directory structure: / workspace / Classification /. A list of files and folders is shown, with '01_LoadData.ipynb' highlighted. A blue arrow points to this file. The main area on the right shows the content of the selected notebook, which includes a section titled '1. Hands-on Deep Learning' and an introduction to the data.

File Edit View Run Kernel Git Tabs Settings Help

Filter files by name

/ workspace / Classification /

Name	Modified
data	2m ago
figures	3m ago
• 01_LoadData.ipynb	55m ago
02_Classification_...	54m ago
03_Interpretability...	54m ago
• 04_KnowledgeDis...	55s ago
09_GradAndAuto...	54m ago
tools.py	3mo ago
TP_Classification_...	3m ago

Terminal 1

01_LoadData.ipynb

Markdown

1. Hands-on Deep Learning

_This hands-on was originally created by Thomas Grenier (T
thomas.grenier@creatis.insa-lyon.fr, michael.sdika@creatis.i

Introduction on data

This hands-on is computer scientist-oriented more than ap

And so, this practice focuses on a meaningless toy example

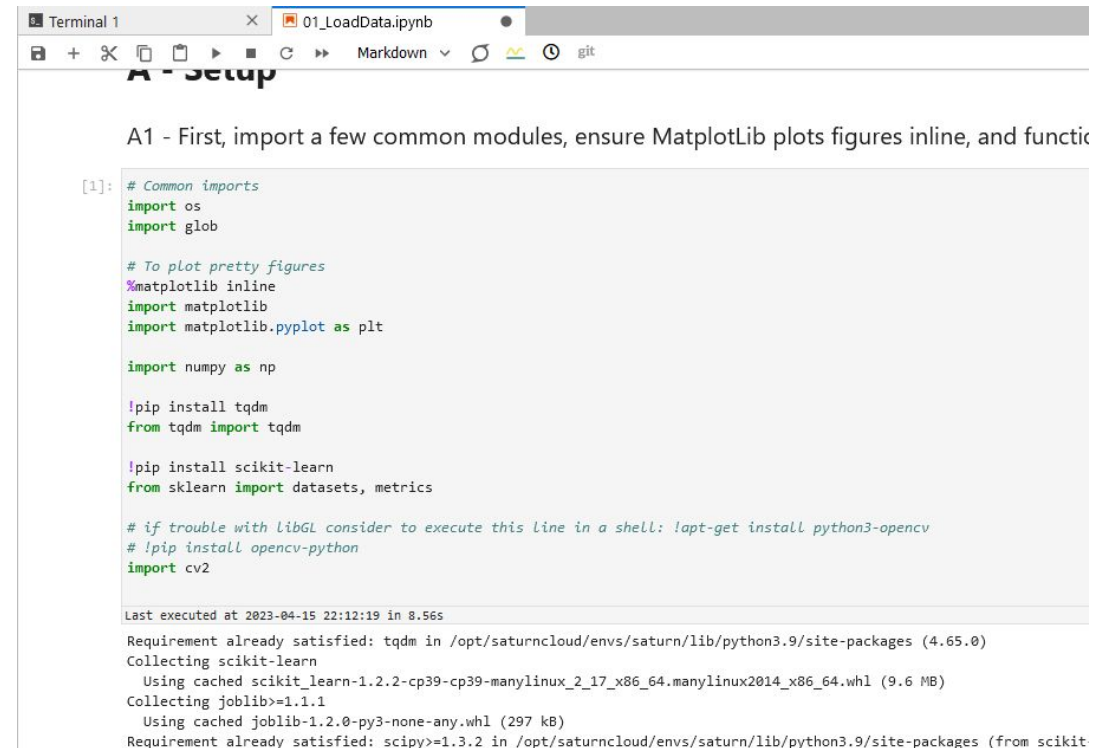
MR images come from [IXI Dataset](#)

Here, we have to recognize whether a brain slice image is a

Avial

Cells and running code

- Notebooks are made of cells : they can contain text or python code
- Python code can be executed from the jupyter lab with the play button 
- Outputs are added in a new cell after the code executed
- A cell can be executed many times
- Generally, a cell requires that the preceding cells have been executed before



The screenshot shows a Jupyter Lab window with a terminal and a code cell. The terminal window at the top shows the command 'Terminal 1' and the file '01_LoadData.ipynb'. The code cell contains Python code for imports and installations. The output of the code cell shows the execution time and the installation of various packages.

```
A1 - First, import a few common modules, ensure Matplotlib plots figures inline, and function
```

```
[1]: # Common imports
import os
import glob

# To plot pretty figures
%matplotlib inline
import matplotlib
import matplotlib.pyplot as plt

import numpy as np

!pip install tqdm
from tqdm import tqdm

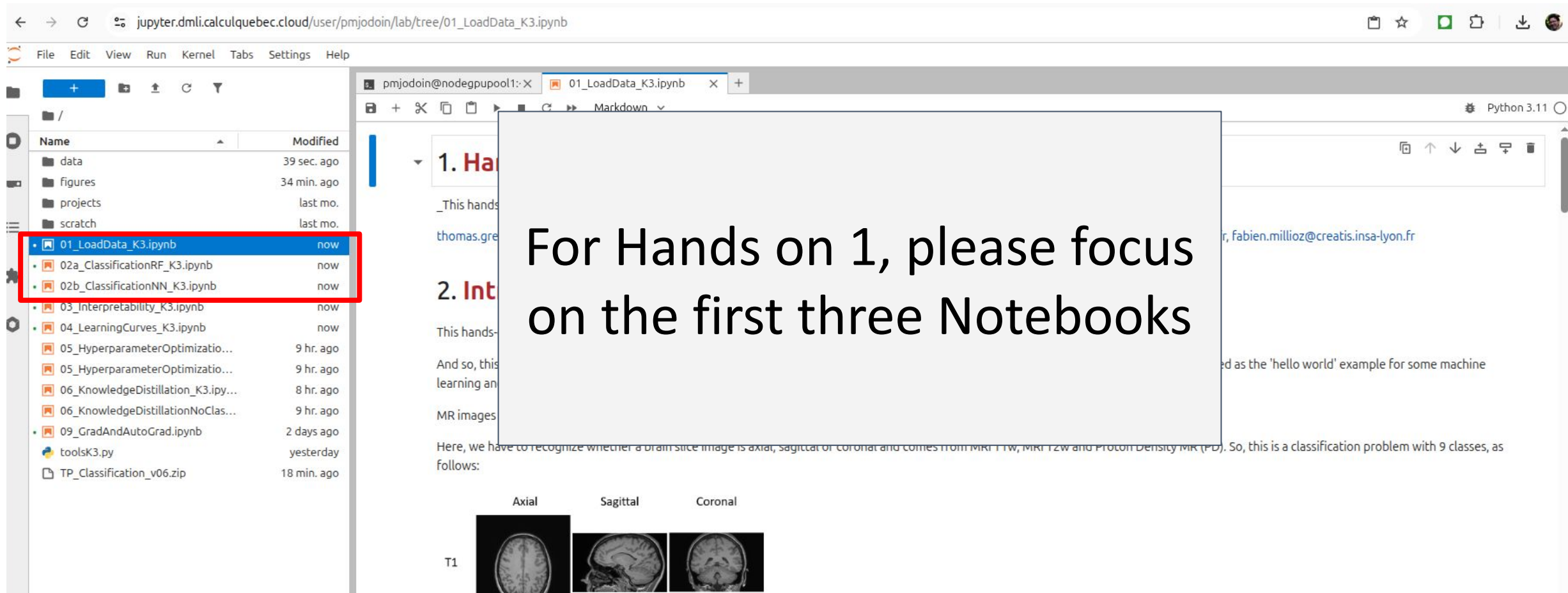
!pip install scikit-learn
from sklearn import datasets, metrics

# if trouble with LibGL consider to execute this line in a shell: !apt-get install python3-opencv
# !pip install opencv-python
import cv2
```

Last executed at 2023-04-15 22:12:19 in 8.56s

Requirement already satisfied: tqdm in /opt/saturncloud/envs/saturn/lib/python3.9/site-packages (4.65.0)
Collecting scikit-learn
Using cached scikit_learn-1.2.2-cp39-cp39-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (9.6 MB)
Collecting joblib>=1.1.1
Using cached joblib-1.2.0-py3-none-any.whl (297 kB)
Requirement already satisfied: scipy>=1.3.2 in /opt/saturncloud/envs/saturn/lib/python3.9/site-packages (from scikit

Hands on 1



For Hands on 1, please focus on the first three Notebooks

1. Hands on 1

2. Introduction

Axial Sagittal Coronal

T1